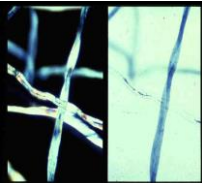




Trace Evidence

- Fibers
- Glass
- Paint Chips
- Hairs
- Glitter
- Soil
- Seeds
- Building Materials
- Feathers
- Tape

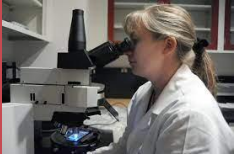


Trace Evidence

Forensic Microscopy
Forensic Geology/Mineralogy

Questions:

- Is this hair human? Where on the body is it from?
- Does the soil on the suspect's shoe match the soil at the crime scene?
- Does the glass on the suspect's clothing have the same physical, chemical, and optical properties as the broken window at the crime scene?



Trace evidence helps solve crimes by linking people, places, and things involved in a crime.

Locard's Principle of Exchange

Whenever two objects come into contact with one another, there is exchange of materials between them.



Lisk/Silva Investigation (1996 – 2002)



Victims: Sofia Silva (16) and sisters Kati (12) and Kristin (15) Lisk



Kara Robinson –
*escaped Evonitz and led
 police to him*

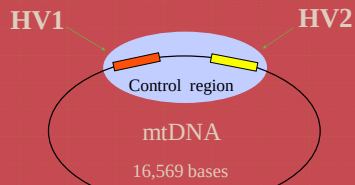


**Trace Evidence Links in the
 Silva/Lisk Investigation
 To Richard Evonitz**

EVIDENCE	SOFA RUM	ARMCHAIR	ADULTS
Head hair microscopically like Evonitz' known hair	X	X	X
Reddish carpet fibers - Evonitz' Frederickshagen residence	X	X	X
Living room carpet fibers - Evonitz' Frederickshagen residence		X	
Carpeting from carpet fibers - Evonitz' Frederickshagen residence	X	X	X
Auto trash fibers from Evonitz' 1992 Ford Taurus	X	X	X
High back rug fibers (South Carolina search)			X
Fibers from Barry Jankovich (South Carolina)	X	X	X
Aluminum fibers (South Carolina search)	X		X
Dark Red acrylic fibers found in Evonitz' known residence and on mattress from Evonitz' home	X	X	X
Light Red acrylic fibers found in garage of white door (South Carolina search)	X	X	X

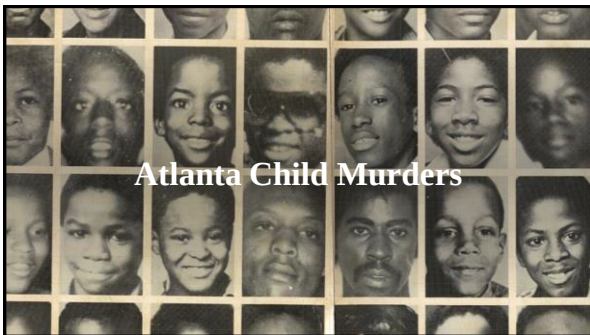


mtDNA match between Evonitz and hair recovered from the victims' bodies



mtDNA first used in 1996





Atlanta Child Murders



Atlanta Child Murders

Trace evidence analysis led to the conviction of Wayne Williams



Paint Analysis



Murder of Tracy Tomko (2000)



Cosmetics: The Murder of Marie Latelle (1912)



Caylee Anthony Murder – Hair Analysis



The Forensic Value of Processed Human Hair Extensions

AKA WEAVOLOGY

A hair from a human derived hair extension recovered from a crime scene

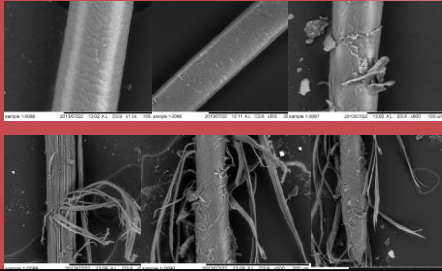


- Identified as a shed telogen human hair
- Analyzed using microscopic techniques and mtDNA
- Sequencing of the mtDNA would place the hair donor, not the suspect, at the scene
- Ability to identify a human hair as an extension might exclude the hair as probative evidence

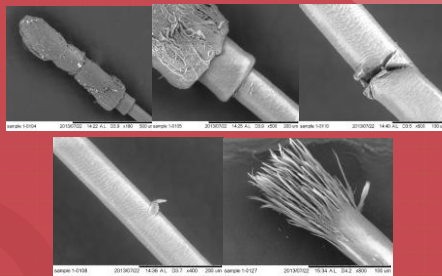
Transmitted Light Microscopy



Scanning Electron Microscopy



Scanning Electron Microscopy –



Conclusion

- **Forensic Implications**
 - microscopic evaluation: features consistent with human hair; cannot definitively classify hair as extension
 - mtDNA analysis: associate donor's maternal lineage with crime
- mtDNA haplotypes: direct correlation between extensions and specific populations not possible
- Multiple donors to a single package of extensions
- Differentiation by chemical analysis is possible – UV-VIS
